DATTA MEGHE COLLEGE OF ENGINEERING, AIROLI
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
STUDENT HIVE OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

SHAIDS

HackHive 2.0

THE HEIST

- TEAM NAME - TRIO.EXE
- TEAM ID -
- DOMAIN - HEALTHCARE
- PROBLEM STATEMENT - OFFLINE-FIRST FIRST-AID ASSISTANCE PLATFORM
- TEAM MEMBER - VAIDEHI BOREKAR, DIVYASHREE BHANGLE, ANUPRAS DAVANE

PROBLEM STATEMENT

In emergency situations such as accidents or natural disasters, reliable internet connectivity is often unavailable, limiting access to digital first-aid resources. Existing first-aid materials are typically text-heavy, complex, and not designed for high-stress scenarios faced by non-medical individuals. The absence of a simple, offline-accessible, and visually guided first-aid solution can delay immediate care and increase the risk of complications before professional medical help arrives.

Target users:

- General public (non-medical individuals)
- Rural communities
- Disaster response volunteers
- Schools and colleges
- Travelers and field workers

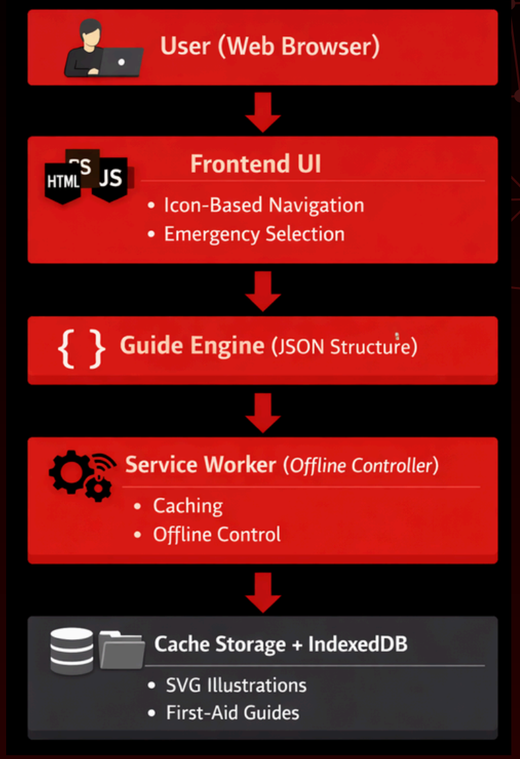
Why the Problem Needs a Solution:

- First 5–10 minutes are critical in emergencies
- Immediate guidance can prevent complications
- Offline accessibility ensures reliability
- Simple visual instructions reduce panic
- Early intervention saves lives before professional help arrives

PROPOSED SOLUTION

How the Solution Can Be Implemented

- Develop a Progressive Web App (PWA)
- Use Service Worker for full offline caching
- Store guides locally using IndexedDB
- Use lightweight SVG illustrations for clarity
- Add push notification & emergency contact shortcut



Practicality

Works offline after first load

No app store dependency

Lightweight & device-compatible

Simple, panic-friendly interface

Sustainability

Built on standardized first aid guidelines

Scalable to more emergencies & languages

Low maintenance cost (web deployment)

Expandable to NGO /government integration

TECHNICAL APPROACH & SYSTEM ARCHITECTURE

APPLICATION LOGIC

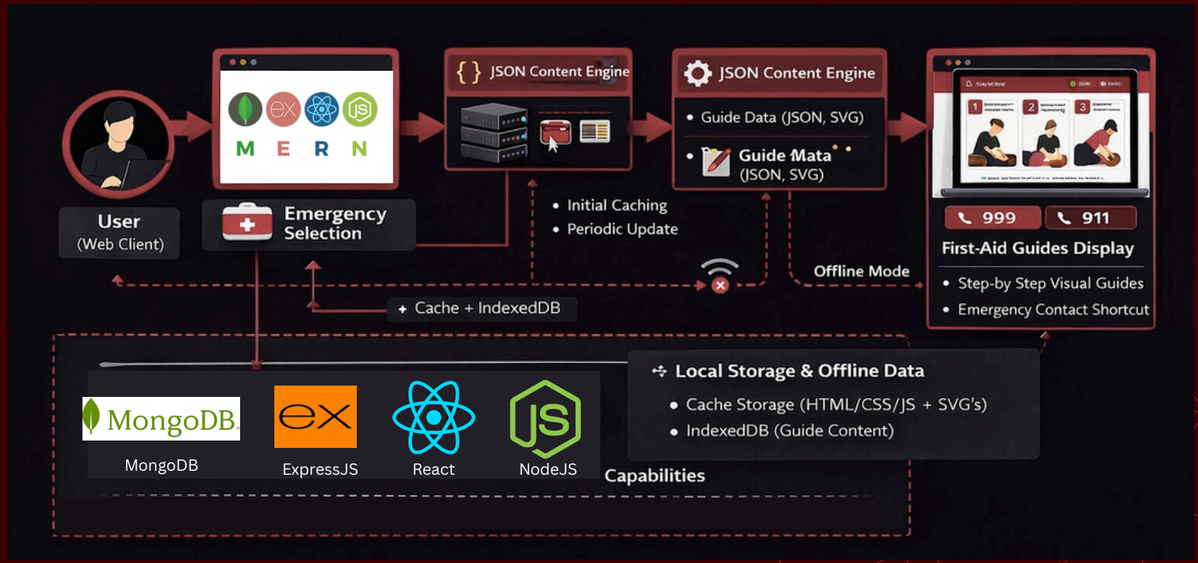
- **TensorFlow.js** – On-device AI emergency classification (text + camera-based recognition)
- **MongoDB + IndexedDB** – Local storage of guides, hospital data & AI resources

FRONTEND

- **React.js (Vite) + Express/Node** – for responsive web interface
- **SVG Illustration Library** – for lightweight visual step guides
- **Web Speech API** – for offline voice guidance & navigation
- **Geolocation API** – for hospital directory & emergency shortcuts

PWA TECHNOLOGIES

- **Service Workers –** Full offline caching of medical guides & assets
- **Cache API –** Store critical content locally
- **Web App Manifest –** Installable app-like experience



FEASIBILITY AND VIABILITY

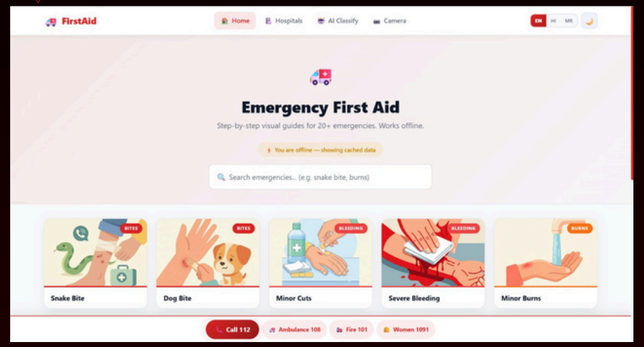
🔧 Feasibility

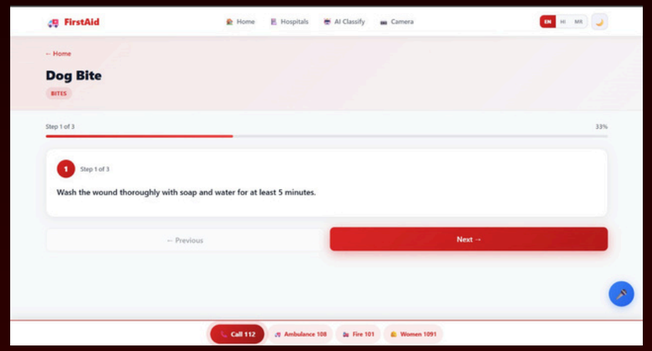
- Develop a Progressive Web App (PWA) using React with Service Worker for full offline functionality.
- Pre-cache all emergency guides, illustrations, and audio during first install.
- Store hospital directory and language packs locally using browser storage.
- Integrate lightweight AI models for text-based classification and offline injury recognition.
- Implement panic-friendly UI with large buttons and icon-based navigation.
- Deploy on secure HTTPS hosting for installability and device API access.

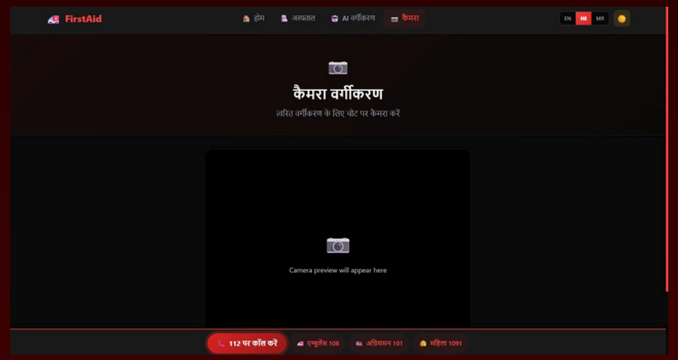
🌍 Viability

- Works without internet, ideal for rural areas and disaster scenarios.
- Runs on any smartphone browser, no need for expensive hardware.
- Lightweight architecture ensures compatibility with low-end devices.
- Minimal maintenance cost (static hosting + local processing).
- Easily scalable to include more emergencies or languages.
- Reduces dependency on constant server infrastructure.

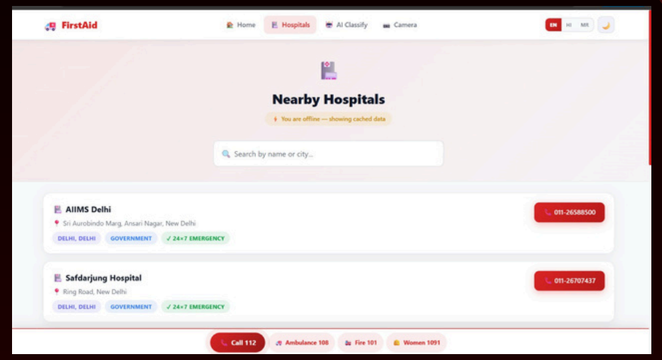
PROTOTYPE

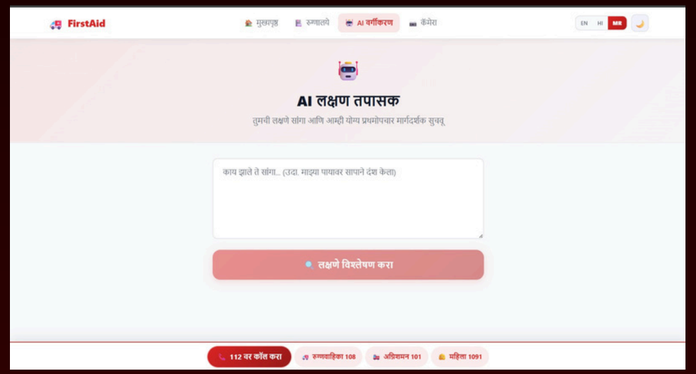






Offline Progressive Web App
Step-by-step visual guides
Icon-based emergency selection
Lightweight SVG illustration library
Emergency contact shortcut
Multi-language support
Panic-friendly large-button UI
Voice-guided instructions (offline audio)
Offline hospital directory
Dark/light mode interface optional
Branching solutions
AI emergency classification (text-based)
Camera-based injury recognition (AI)
- Voice Navigation





REFERENCES

- https://www.who.int/ -World Health Organization (WHO)
- https://react.dev/ -React Documentation
- https://developer.mozilla.org/en-US/ -Service Worker API (MDN)
- https://www.mayoclinic.org/first-aid/ -Mayo Clinic First Aid Guide
- https://web.dev/progressive-web-apps/ -Google Developers – PWA Guide
- https://www.india.gov.in -Government of India – Emergency Helpline Directory